

Sarang Shigwan

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Summary

AI Full Stack Engineer with experience building AI-native applications using RAG pipelines, vector embeddings, and cloud-native architectures. Skilled in developing scalable APIs, full-stack systems, and deploying production-ready solutions on AWS using Docker and CI/CD. Built real-world systems for failure analysis, document intelligence, and observability. Passionate about solving real-world problems using AI-driven engineering.

Education

- **B.E. ENTC** at Marathwada Mitra Mandal's College of Engineering, Pune
Nov 2022 – May 2026 | **CGPA: 9.24 / 10**
- **H.S.C.** at Modern College of Arts, Science & Commerce, Pune
2020 – 2022 | **74.17%**
- **S.S.C.** at St. Francis De Sales School, Pune
2007 – 2020 | **89.80%**

Experience

Backend Intern(AWS Cloud) - EvolvingX Pvt. Ltd.

Oct 2024 – June 2025

- Developed a full-stack web application (React + Django) with REST APIs supporting dynamic user workflows
- Deployed production system on AWS EC2 with HTTPS and database integration
- Improved backend efficiency by optimizing API handling and middleware design
- Ensured secure communication using token-based authentication.

Projects

LexVault (Smart Legal Document Manager)

Github Repo: [Lexvault](#)

- Built a document intelligence system with version tracking and audit history
- Designed PostgreSQL schema for efficient retrieval of document versions
- Developed REST APIs + React dashboard
- Designed foundation for AI-based document search using embeddings.

Skills : FastAPI, PostgreSQL, SQL, REST API Design, Database Schema Design, Version Control Systems, Data Modeling.

TraceLens

Github Repo: [Tracelens](#) | **Design Report :** [Report](#)

- Built an AI-powered root cause analysis system using a RAG pipeline with vector embeddings to analyze distributed system failures
- Designed a FastAPI-based microservice deployed on AWS Lambda for scalable, stateless inference
- Implemented vector similarity search with cosine scoring to identify failure patterns with confidence scoring
- Enabled automated debugging workflows, reducing manual log analysis effort.

Skills : Sentence-Transformers (all-MiniLM-L6-v2), RAG, NumPy, Pydantic, Vector Similarity Search, Cosine Similarity, NLP Embeddings

HFT Latency Monitoring Platform

Github Repo: [LatencyMonitor](#) | **Design Report :** [Report](#)

- Built a sub-millisecond UDP ingestion pipeline with P50/P95/P99 latency tracking on Kubernetes.
- Implemented Prometheus histograms and Grafana dashboards for real-time performance observability.
- Containerized a multi-service Go architecture using Docker and Kubernetes, debugging runtime issues for reliable deployments.

Skills : Low-latency systems, Go concurrency, Docker, Kubernetes, Prometheus/Grafana, Linux debugging, UDP pipelines.

Skills

- **Fundamentals :** Operating Systems, DBMS, Networking, OOP, DSA, SDLC, system design
- **GenAI :** RAG Pipelines, LLM Integration, Prompt Engineering, Vector Embeddings, Semantic Search, AI Workflows, Structured Outputs, NLP (Sentence Transformers), AI-assisted Development (GitHub Copilot, Claude)
- **Fullstack :** Python (FastAPI, Django), REST API Design, Microservices Architecture, React.js, Authentication (JWT), Database Design (PostgreSQL)
- **Cloud/DevOps :** AWS (EC2, S3, Lambda, IAM, VPC), Docker, Terraform, CI/CD (GitHub Actions), Cloud Deployment, Infrastructure as Code, Monitoring (Prometheus, Grafana, CloudWatch)

Achievements

- **AIR 2** SAENIS Efficycle 2024
- **Winner** at Hackathons like Udaan 2023 and Dextirity 2024
- **Research paper published on IEEE Explore :** <https://ieeexplore.ieee.org/document/11447964>